Providing a Method for Hiding Sensitive Smartphone Data Using AES/GCM Encryption

Master of Science Thesis

Naeemeh Etesami

Date: Summer 2018

Abstract

The proliferation of mobile cloud computing has transformed smartphones into critical repositories of personal and professional data, making them high-value targets for cyber threats. While traditional cryptography ensures data confidentiality, the mere presence of encrypted traffic creates a metadata signature that can attract targeted attacks. This research addresses this vulnerability by proposing a comprehensive dual-layer security framework designed specifically for mobile environments.

The first layer utilizes the Advanced Encryption Standard (AES) in Galois/Counter Mode (GCM) to provide robust confidentiality, data integrity, and authentication. The second layer introduces a novel "Transparent Steganography" algorithm. Unlike traditional methods that leave visual artifacts, this technique embeds encrypted data into standard text messages using zero-width Unicode characters, rendering the hidden communication mathematically and visually imperceptible to human observers and standard text parsers.

To validate the feasibility of the cryptographic component on resource-constrained devices, a non-pipelined hardware architecture for an AES-based core (JH hash function) was designed and simulated using VHDL. Simulation results demonstrated a throughput of approximately 664 Mbps, confirming the design's efficiency. The study concludes that combining authenticated encryption with transparent steganography provides a viable, high-security solution for protecting sensitive smartphone data against both interception and detection.

List of Abbreviations

AEAD	Authenticated Encryption with Associated Data
AES	Advanced Encryption Standard
CIA	Confidentiality, Integrity, and Availability
CM	Cover Message
CTR	Counter Mode
DES	Data Encryption Standard
DoS	Denial of Service
FSM	Finite State Machine
GCM	Galois/Counter Mode
HM	Hidden Message
ISE	Integrated Synthesis Environment (Xilinx)
LSB	Least Significant Bit
MITM	Man-in-the-Middle
NIST	National Institute of Standards and Technology
SPN	Substitution-Permutation Network
VHDL	VHSIC Hardware Description Language

Electrical and Computer Engineering
Islamic Azad University, Science and Research Branch Faculty of

Title:	Providing a Method for Hiding Sensitive Smartphone Data Using AES/GCM Encryption
Author:	Naeemeh Etesami
Supervisor:	Seyed Hossein Erfani, Ph.D.
Project Type:	Master of Science Thesis
Project Period:	1 March 2018-15 September 2018
Location:	Islamic Azad University, Science and Research Branch Faculty of Electrical and Computer Engineering

Contents

Chapter 1: Introduction 1

1.1. Introduction and Literature Review 1

1.2. Problem Statement 1

1.3. Research Objectives 1

Chapter 2: Proposed Methodology 3

2.1. Framework Overview and Research Assumptions 3

2.2. Problem Inputs and System Architecture 3

2.3. Transparent Steganography and Data Flow 3

Chapter 3: Implementation and Experimental Results 5

3.1. Implementation Overview and Hardware Architecture 5

3.2. Simulation Environment and Execution 5

3.3. Cryptographic Performance Analysis 5

3.4. Steganographic Analysis 6

3.5. Network Feasibility Analysis 6

Chapter 4: Results, Analysis, and Limitations 7

4.1. Steganographic Analysis: Imperceptibility 7

4.2. Cryptographic Analysis: Throughput and Efficiency 7

4.3. Network Feasibility Analysis (Opportunistic Communication) 7

4.4. Overall Security Assessment 8

4.5. Limitations 8

Chapter 5: Conclusion and Future Directions 9

5.1. Summary of Key Findings 9

5.2. Contribution of the Work 9

5.3. Recommendation for Future Research 9

Chapter 6: References 11

Chapter 7: Appendices 13

Appendix A: Steganography Algorithm Logic 13

Appendix B: Cryptographic Core VHDL Specification 14

Chapter 1: Introduction

1.1. Introduction and Literature Review

The evolution of computational paradigms has been profoundly influenced by the proliferation of mobile and cloud computing. Smartphones, in particular, have transitioned from simple communication tools to indispensable hubs of personal and professional data. They host sensitive information, including GPS locations, medical data, financial credentials, and user profiles. While this data enables powerful applications in fields like personal health monitoring and intelligent transport systems, it also creates a high-value target for adversaries.

As data processing moves to cloud models like PaaS, SaaS, and IaaS, the security of data in transit and at rest on mobile database servers becomes a critical factor for user adoption. Information security management is imperative to preserve confidentiality, integrity, and availability (CIA). However, traditional security measures often focus on securing the server or the network, while the mobile device itself, and the data it transmits, remains a vulnerable link. The portable nature of these devices exposes them to unique threats, including physical theft, data recovery from physical memory, malware, and phishing attacks.

1.2. Problem Statement

The standard response to data sensitivity is encryption. Cryptography provides robust symmetric and asymmetric algorithms to protect data confidentiality. However, encryption alone is not a panacea. The very presence of an encrypted data stream, while its content is unreadable, acts as a clear metadata signature. It signals to an adversary that a communication of value is occurring, inviting attacks such as targeted decryption attempts, Man-in-the-Middle (MITM) interceptions, or Denial of Service (DoS) attacks where the adversary simply blocks the transmission.

This research addresses a fundamental flaw: encryption conceals the *content* of a message, but it does not conceal the *existence* of the message. For truly sensitive data, a higher level of security is required—one that achieves "security by obscurity" not as a replacement for strong cryptography, but as a complementary layer. This is the domain of steganography. However, traditional text-based steganography is often rudimentary; methods using special formatting or extra spaces are visually detectable and easily broken by simple text analysis. Therefore, a dual-layer approach is needed: robust encryption paired with a novel, imperceptible steganography technique.

1.3. Research Objectives

This research aims to design, implement, and analyze a complete framework for securing sensitive smartphone data. The primary objectives are:

1. **To analyze and select a robust cryptographic protocol** suitable for mobile environments, focusing on the AES/GCM standard for its provision of confidentiality, integrity, and authentication.
2. **To design and develop a novel text steganography method** that is transparent, imperceptible, and resistant to automated analysis, overcoming the flaws of existing methods.
3. **To integrate these two layers** into a single, cohesive security framework.
4. **To design and simulate a hardware architecture** for the AES-based cryptographic core to validate its performance and resource requirements.

The primary contribution of this work is a novel "transparent steganography" algorithm that utilizes zero-width Unicode characters to embed data in any text-based medium without any visual alteration. This, combined with a VHDL-based simulation of an AES-like architecture, provides a validated, high-security solution for sensitive data.

Chapter 2: Proposed Methodology

2.1. Framework Overview and Research Assumptions

This chapter details the proposed dual-layer security framework designed to protect sensitive data on smartphones. Unlike traditional approaches that rely solely on encryption, this research integrates robust cryptography with a novel, transparent steganography method. The methodology is specifically engineered to address the constraints of mobile environments, such as limited processing power and the necessity for covert communication over standard text-based channels like SMS or social media. To define the operational scope, the system functions under the premise that the communication channel—whether cellular or Wi-Fi—is inherently insecure and monitored by adversaries capable of intercepting, analyzing, and potentially blocking traffic. Furthermore, given the battery and computational limitations of smartphones, the design prioritizes efficiency over computationally expensive asymmetric schemes. It is also established that the sender and receiver have securely exchanged a symmetric key prior to transmission, as the primary focus remains on secure transmission and hiding rather than key exchange. Finally, the cover medium is restricted to ubiquitous text formats such as SMS, email, or chat logs to ensure broad compatibility and camouflage.

2.2. Problem Inputs and System Architecture

The proposed algorithm operates on three fundamental inputs: the Hidden Message (HM), the Cover Message (CM), and the Secret Key (K). The Hidden Message comprises the sensitive information, such as GPS coordinates or passwords, that the user intends to transmit securely. The Cover Message is a standard, innocuous text string—for example, "Hello, I am on my way"—which acts as the carrier and remains visible to any observer. The Secret Key is a 128, 192, or 256-bit symmetric key utilized by the encryption algorithm to protect the payload. The solution implements a sequential pipeline consisting of two distinct engines: the Cryptographic Engine and the Steganographic Engine. The first layer ensures the confidentiality and integrity of the Hidden Message through the Advanced Encryption Standard (AES) in Galois/Counter Mode (GCM). GCM was selected specifically for its provision of Authenticated Encryption with Associated Data (AEAD). By utilizing Counter Mode (CTR), it allows for high-speed, parallelizable encryption suitable for mobile processors, while simultaneously computing a GHASH tag to ensure data integrity. In this process, the message is encrypted using the secret key to produce the ciphertext (C) and an authentication tag (T), resulting in an encrypted payload defined as the combination of these two elements.

2.3. Transparent Steganography and Data Flow

The second layer embeds the encrypted payload into the Cover Message using a novel approach. Unlike traditional text steganography techniques that alter spacing or fonts—methods easily detected by automated analysis—this framework employs an innovation based on Zero-Width Unicode characters. These non-printing characters, such as U+288B and U+288C, possess no visual width, rendering them mathematically present but visually invisible to the human eye. The encoding strategy maps the binary bits of the encrypted payload to these specific characters; for instance, a binary '0' is mapped to U+288B, and a '1' is mapped to U+288C. To further increase entropy and resist pattern analysis, a reversible scrambling function is applied to the binary string prior to embedding. This function inverts bits based on the string length to prevent statistical detection of the hidden data.

The complete operational flow proceeds sequentially from the sender to the receiver. On the sender's side, the process begins with the reception of the hidden message, cover message, and key. The system applies AES/GCM to the hidden message to generate the encrypted payload, which is subsequently converted into a raw binary string. This binary data undergoes the scrambling function to enhance security before being translated into the corresponding string of zero-width Unicode characters. Finally, this invisible Unicode string is injected by appending it to the cover message to create the final Transmitted Message. Upon reaching the receiver, the decoding process initiates by scanning the incoming text for zero-width characters to reconstruct the scrambled binary string. Once extracted, the inverse scrambling function is applied to restore the original binary sequence. This data is then passed to the AES/GCM engine, which verifies the authentication tag (T). If the tag is valid, the engine decrypts the ciphertext (C) to reveal the original hidden message, ensuring that the data has remained confidential and tamper-proof throughout transmission.

Chapter 3: Implementation and Experimental Results

3.1. Implementation Overview and Hardware Architecture

This chapter presents the implementation details of the proposed framework and analyzes the data extracted from the simulation experiments. The evaluation focuses on two key areas: the hardware performance of the cryptographic core (AES/JH architecture) and the qualitative analysis of the steganographic engine. To validate the cryptographic layer, a hardware architecture was designed and implemented using VHDL. The core design utilizes the JH hash function architecture, which shares the Substitution-Permutation Network (SPN) structure with AES, serving as a robust proxy for performance analysis.

In the architectural design, a non-pipelined approach was selected to prioritize area efficiency, a critical constraint for mobile devices with limited silicon real estate. In this configuration, the 1024-bit state is processed iteratively through a 42-round loop controlled by a feedback mechanism, rather than unrolling the rounds into a larger, resource-intensive pipeline. To orchestrate the complex data flow, a 5-state Finite State Machine (FSM) was implemented. This control logic manages the system's operation through distinct phases, transitioning from Initialization (S0) to the main Processing loop (S2), and finally to the Finalization state (S5) upon completion of the cryptographic rounds.

3.2. Simulation Environment and Execution

The hardware design was synthesized and simulated using the Xilinx ISE Design Suite 13.2, utilizing the ISim simulator for verification. A comprehensive VHDL test bench was constructed to validate the design, serving as a harness to feed known message vectors (mes_in) into the system while monitoring the resulting output hash (hash_out) and completion signals (ack_out). The simulation execution was governed by a clock set to a period of 10ns. Under these parameters, the system executed the full 42 rounds of the cryptographic function for each data block, allowing for precise measurement of timing and logic behavior.

3.3. Cryptographic Performance Analysis

Upon running the simulation, quantitative data was extracted to determine the efficiency of the cryptographic engine. The first phase of analysis involved functional verification through the examination of Register Transfer Level (RTL) schematics and waveforms. This process confirmed the functional correctness of the design, as the FINISHED_42_ROUNDS signal successfully asserted to '1' immediately following the completion of the processing cycles, producing the correct hash digest for the given input vector.

The primary performance metric extracted from the simulation was throughput, which indicates the speed at which the mobile device can encrypt data. For a single 512-bit message block, the process required a total of 77 clock cycles. Given the 10ns clock period, the total computation time was recorded as 770 nanoseconds. Based on this timing data, the throughput is calculated as follows:

$$\mathbf{Throughput} = \mathbf{MessageSize/TotalTime} = (\mathbf{512}\ \mathbf{"bits"})/(\mathbf{770}\ \mathbf{"ns"}) \approx \mathbf{664}\ \mathbf{"Mbps}$$

This analysis confirms that the throughput of approximately 664 Mbps provided by the non-pipelined architecture offers sufficient speed for real-time mobile data encryption. Furthermore, it achieves this performance while maintaining a compact hardware footprint, validating the design choices made for resource-constrained mobile environments.

3.4. Steganographic Analysis

The output of the steganographic engine was analyzed to verify the "imperceptibility" of the hidden data. This involved a visual inspection where the Transmitted Message (comprising the Cover Message and the Hidden Payload) was compared directly against the original Cover Message. The result showed that the two text strings were visually identical. The utilization of zero-width Unicode characters (U+288B and U+288C) resulted in zero visual artifacts, effectively rendering the hidden payload invisible.

This method stands in contrast to traditional steganographic techniques. Unlike format-based steganography, which introduces detectable extra spaces, or symbol-based methods that rely on visible emoticons, the data extracted in this framework remains invisible to both human readers and standard text parsers. This qualitative assessment confirms the successful implementation of "Transparent Steganography," ensuring that the existence of the secure communication channel remains concealed.

3.5. Network Feasibility Analysis

To ensure the framework is viable in real-world scenarios, an opportunistic communication experiment was conducted using Nokia N900 smartphones. The scenario involved exchanging data via Bluetooth between a stationary device and a moving device to simulate a brief encounter. In a contact window of approximately 30 seconds, the devices successfully transferred an average of 563.25 KB of data. This result leads to the conclusion that the available bandwidth in such opportunistic scenarios is more than sufficient to transfer the text-based encrypted payloads generated by the proposed framework.

Chapter 4: Results, Analysis, and Limitations

This chapter presents a comprehensive analysis of the proposed dual-layer security framework. The evaluation is systematically divided into three main components: a qualitative assessment of the transparent steganography method, a quantitative performance analysis of the cryptographic core, and a feasibility study regarding opportunistic data transfer. Finally, the limitations of the current research implementation are discussed to provide a balanced and objective view of the system's capabilities and constraints.

4.1. Steganographic Analysis: Imperceptibility

The primary contribution of this research is the "Transparent Steganography" algorithm, and its performance was evaluated by comparing the visual output of the proposed method against traditional text-hiding techniques. Standard text steganography often relies on modifying font properties, inserting visible symbols, or altering spacing, such as using double spaces. These methods inevitably create visual anomalies that are easily detectable by human readers or basic text-analysis algorithms, thereby compromising the secrecy of the communication.

In contrast, the proposed algorithm utilizes zero-width Unicode characters, specifically U+288B and U+288C, to encode binary data. Experimental results demonstrated that a Cover Message (CM) containing a Hidden Message (HM) is visually indistinguishable from the original text. By shifting the hiding mechanism from the visual layer to the character-encoding layer, the method successfully achieves "security by obscurity." This ensures that the existence of the message remains concealed from third-party observers, offering a significant advantage over legacy techniques.

4.2. Cryptographic Analysis: Throughput and Efficiency

The hardware performance of the cryptographic engine, based on the AES/JH architecture, was evaluated using the VHDL simulation results detailed in the previous chapter. Functional verification confirmed that the 42-round core architecture correctly processes input vectors and produces the expected hash digest, thereby validating the logic of the Finite State Machine (FSM) and the Substitution-Permutation Network (SPN).

Regarding quantitative metrics, the simulation indicated that the encryption of a single 512-bit block required 770 nanoseconds, corresponding to 77 clock cycles at a 10ns period. Consequently, the calculated data rate for this non-pipeline architecture is approximately 664 Mbps. While fully pipelined architectures can typically achieve higher speeds, this analysis confirms that a compact, non-pipelined core is sufficient for real-time encryption on mobile devices, where minimizing silicon area and power consumption are critical design constraints.

4.3. Network Feasibility Analysis (Opportunistic Communication)

To validate the practicality of exchanging these secure messages within a real-world mobile environment, an opportunistic Bluetooth communication experiment was conducted. The experimental setup utilized two Nokia N900 smartphones to simulate a "contact window" between a stationary sender and a moving receiver. A 30-second inquiry interval resulted in a 100% device discovery rate, establishing a reliable baseline for connection.

In terms of transfer capacity, the devices successfully exchanged an average of 563.25 KB of data within an approximate 12-second window, with a peak transfer recording of 1,517.58 KB. Given that text-based steganographic messages are typically small—often in the kilobyte range—this experiment proves that even brief, opportunistic encounters provide ample bandwidth to securely exchange encryption keys or hidden messages.

4.4. Overall Security Assessment

The integration of these layers provides a robust defense-in-depth solution. The AES/GCM layer ensures confidentiality and integrity, guaranteeing that even if the data is intercepted, it cannot be read or tampered with without the correct key. Simultaneously, the steganography layer ensures covertness, making the communication channel appear benign and effectively preventing traffic analysis attacks.

4.5. Limitations

While the proposed framework successfully meets its primary research objectives, several limitations were identified during the study. First, a hardware architecture trade-off was made by implementing a non-pipeline architecture for the cryptographic core. While this minimizes the hardware area, making it suitable for small chips, it limits the maximum potential throughput compared to a fully pipelined design. Second, the scrambling function used in the steganography layer, which involves inverting bits based on string length, is relatively simplistic. Although effective for obscuring patterns, it is not a cryptographically secure pseudo-random number generator (CSPRNG).

Furthermore, the research validated the core components separately—using VHDL for the hardware simulation and Python scripts for the steganography—meaning a fully integrated, user-friendly mobile application, such as an Android keyboard app, was not developed as part of this thesis. Finally, the opportunistic communication model relies on physical proximity due to Bluetooth range limitations, which restricts the use case primarily to face-to-face or close-range data exchange scenarios.

Chapter 5: Conclusion and Future Directions

5.1. Summary of Key Findings

This research aimed to develop a robust solution for securing sensitive smartphone data by addressing both confidentiality and covertness. The experimental validation of the proposed dual-layer framework yielded several significant findings regarding the system's performance and viability. Qualitatively, the analysis demonstrated that the proposed "Transparent Steganography" method completely eliminates visual artifacts. By mapping encrypted bits to zero-width Unicode characters (e.g., U+288B and U+288C), the hidden message remains visually indistinguishable from the cover text, successfully achieving the goal of "security by obscurity".

Quantitatively, the hardware simulation of the AES-based JH cryptographic core confirmed its suitability for mobile environments. The non-pipelined architecture achieved a throughput of approximately 664 Mbps with a total computation time of 770 nanoseconds for a 512-bit block. This metric proves that high-security encryption can be implemented efficiently on resource-constrained hardware without compromising speed. Furthermore, the operational feasibility was verified through opportunistic network analysis. Even in brief contact windows of under 20 seconds, the system was capable of transferring an average of 563.25 KB of data. This capacity is more than sufficient for exchanging the text-based payloads generated by this framework, validating its practicality for real-world scenarios.

5.2. Contribution of the Work

This thesis contributes to the field of mobile information security through three primary innovations. First, it introduces a novel transparent steganography algorithm. Unlike traditional methods that rely on detectable formatting changes, such as double spaces or emoticons, this work utilizes non-printing Unicode characters. This approach renders the existence of the message invisible to both human observers and standard text parsers, significantly advancing the state of the art in text hiding.

Second, the research presents an integrated dual-layer security framework. By successfully combining the cryptographic strength of AES/GCM—which provides confidentiality, integrity, and authentication—with the stealth of transparent steganography, this holistic approach offers a comprehensive defense. It protects data from interception while simultaneously mitigating the risk of traffic analysis by hiding the metadata signature typically associated with encryption.

Third, the thesis provides hardware-level validation of the cryptographic component. By designing and simulating the VHDL architecture for the cryptographic core, this work offers concrete evidence of the system's performance characteristics, specifically throughput and area efficiency. This effectively bridges the gap between theoretical algorithm design and practical hardware implementation for mobile devices.

5.3. Recommendation for Future Research

To further advance the capabilities of this security framework, several directions are recommended for future investigation. A critical next step is the implementation of a pipeline architecture. While the current non-pipeline design prioritizes small silicon area, future work should focus on unrolling the 42 rounds into distinct hardware

stages. This modification would significantly enhance throughput for high-bandwidth applications, albeit at the cost of increased chip area.

Additionally, the research suggests extending the system to support higher security standards. While the current simulation validated the 256-bit architecture, extending the core to support JH-512 and JH-1024 would provide greater resistance against future threats, including those posed by quantum computing attacks. Finally, the ultimate step toward practical adoption is the development of a user-facing mobile application. Future efforts should focus on creating an app for Android or iOS that integrates the cryptographic and steganographic engines into a custom keyboard or messaging client. This would allow for seamless, user-friendly secure communication, moving the technology from a research prototype to a consumer-ready privacy tool.

Chapter 6: References

- [1] Abed, Elham, Ebrahimi Atani, Reza, "A review of botnets on smartphones and countermeasures," *Monadi Security of Production and Exchange of Information*, p. 7, 1392 (2013).
- [2] F. Mohammadreza, "The role of information security management in reducing the operational risk of financial institutions," 1391 (2012).
- [3] Sarabi, Mohammad, Izadi Roodmajani, Ali Akbar, Salehan, Alireza, "Investigating information security methods in mobile phones," *International Conference on E-Citizen and Mobile Phone*, 1389 (2010).
- [4] Bala-far, Mohammad Ali, Razavi, Seyed Naser, Memarbashi, Elham, "A new image steganography algorithm based on the honey bee optimization method," *Master's thesis for the degree of M.Sc. in Information Technology Engineering, Secure Telecommunications specialization*, 1395 (2016).
- [5] Delforouzi, A. et al., "Providing a lossless information steganography method in digital audio based on progressive wavelet transform," *National Conference of the Computer Society of Iran*, 1386 (2007).
- [6] Naghsh-Nilchi, A. et al., "A new method in robust data steganography in JPEG images," *Journal of Electrical and Computer Engineering of Iran*, 1385 (2006).
- [7] Javadi, Seyed Mohammad Ali, Hassanzadeh, Maryam, "The effect of color spaces on information hiding in color images," *Scientific-Research Journal of Soft Computing and Information Technology*, Vol. 4, No. 3, 1394 (2015).
- [8] Bayat, Manouchehr; Ahmad Hassannia & Mohammad Tahghighi Sharbian, "Investigation and comparison of cryptography of symmetric algorithms and asymmetric algorithms," *First Regional Conference on Theoretical and Applied Research in Computer Engineering and Information Technology, Saqqez, University of Applied Science and Technology of Saqqez*, 1394 (2015).
- [9] Kaffash, Ali & Javad Rajabnia, "Investigation and comparison of the performance of symmetric and asymmetric encryption algorithms," *International Conference on Computer Engineering and Information Technology, Tehran, Permanent Secretariat of the Conference*, 1395 (2016).
- [10] Sadeghzadeh, Seyed Hassan, Danesh Arasteh, Anseh, Gholamzadeh Moghaddam, Mohsen, "Increasing security in mobile payment systems (mobile banking) using a cryptographic system," 1390 (2011).
- [11] Karimanpour, Hamidreza, Pooyan, Seyed Ali, "Increasing security in the communications of power dispatching centers by using advanced information encryption algorithms with a case study of the AES algorithm," *30th International Power System Conference*, 1394 (2015).
- [12] Masoumi, Mehdi, Masoumi, Masoud, Ahmadian, Mohammad, "Practical implementation of differential power analysis on the AES encryption system," *7th International Conference of the Iranian Society of Cryptology, K. N. Toosi University of Technology*, 1389 (2010).
- [13] Rezayati, Mohammad Hadi, Amin, Ahmad Reza, Yousefi, Hamed, "A practical and effective power analysis attack against the hardware implementation of the Advanced Encryption Standard system," *Electrical Engineering and Sustainable Development, with a focus on new achievements in electrical engineering*, 1392 (2013).

- [14] Hemmati, Sara, Taghipour Eivazi, Shiva, “A review of encryption methods and algorithms,” *First International Comprehensive Conference on Engineering Sciences in Iran*, 1395 (2016).
- [15] Ghoochani, Samira, Saeedi, Shahram, “Encryption of confidential information for steganography in digital images,” *First International Comprehensive Conference on Engineering Sciences in Iran*, 1395 (2016).
- [16] Kowsarifard, Seyed Reza; Ali Rafiei & Hassan Masoumi, “Data hiding in digital media using frequency domain processing,” *Second National Conference on New Technologies in Electrical and Robotics Industries, Eslamshahr, Sama Technical and Vocational College, Eslamshahr Branch*, 1393 (2014).
- [17] Ghorashi, Hassan & Negar Ghani, “Data hiding in audio signals,” *6th Iranian Conference on Electrical and Electronics Engineering, Gonabad, Islamic Azad University, Gonabad Branch*, 1393 (2014).
- [18] Puthal, D., N., S., R., R., & C., J., "DPBSV--An efficient and secure scheme for big sensing data stream." In *Trustcom/BigDataSE/ISPA, IEEE*, vol. 1, pp. 246-253, (2016).
- [19] Mushtaque, M. A., Dhiman, H., Hussain, S., & Maheshwari, S., "Evaluation of DES, TDES, AES, blowfish and two fish encryption algorithm: based on space complexity," *International Journal of Engineering Research & Technology (IJERT)*, (2014).
- [20] Klasnja, P., Consolvo, S., Choudhury, T., Beckwith, R., & Hightower, J., "Exploring privacy concerns about personal sensing," *Pervasive Computing*, pp. 176-183, (2009).
- [21] Chi-Kwong Chan, L.M. Cheng, "Hiding data in images by simple LSB substitution," *Department of Computer Engineering and Information Technology, City University of Hong Kong, Hong Kong*, (2003).
- [22] "ADVANCED ENCRYPTION STANDARD (AES)," *Federal Information Processing Standards Publication 197*, (2001).
- [23] Sakalli, M. T., "Cryptography Education for Students,” *Information Technology Based Higher Education and Training, Proceedings of the Fifth International Conference on*, pp. 621 - 626, (2001).
- [24] Gurmann, Peter, Naccache, David, and Palmer, C., "What Is Cryptography," *IEEE Computer Society*, (2006).
- [25] Stinson, D. R., "Cryptography, Theory and Practice," *Second edition, Chapman & Hall/CRC Press*, (2002).
- [26] Li Xi and Hu Han, Ping, "Efficient of secure mobile payment," *Institute for pattern recognition and artificial Intelligence, Huazhong university science and technology china, journal of communication*, (2007).
- [27] Luo, C., Fylakis, A., Partala, J., Klakegg, S., Goncalves, J., Liang, K., ... & Kostakos, V., "A data hiding approach for sensitive smartphone data," *In Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, pp. 557-468, (2016).
- [28] Nai Miao, Yutao He and Jane Dong, "hymnMark: Towards Efficient Digital Watermarking on Android Smartphones," *In Proceedings of the International Conference on Wireless Networks (ICWN)*, pp. 1-8, (2012).
- [29] Hsueh, C., Lin, J.Y., & Lin, M.Y., "Secure cloud storage for conventional data archive of smart phones," IEEE Int. Symposium on Consumer Electronics, ISCE’11, Singapore, (2011).
- [30] Jia, W., Zhu, H., Cao, Z., Wei, L., & Li, X., "A secure data service mechanism in mobile cloud computing," *IEEE Conference on Computer Communications Workshops, INFOCOM WKSHPS*, (2011).

- [31] Sahai, A., "Secure Protocols for Complex Tasks in Complex Environments," *In INDOCRYPT*, pp. 14-16, (2004).

Chapter 7: Appendices

Appendix A: Steganography Algorithm Logic

This appendix provides the implementation logic for the "Transparent Steganography" engine, specifically the scrambling and Unicode injection phases described in Chapter 3.

A.1. Scrambling Function Logic

The following pseudo-code implements the reversible scrambling algorithm used to increase entropy before embedding.

Python

```
def scramble_data(binary_string):
    """
    Implements the scrambling logic defined in Section 3.3.4.
    """
    T = len(binary_string)

    # Calculate Multiplier M based on length [cite: 226, 227]
    if (T % 2) == 0:
        M = 2  # Even length
    else:
        M = 3  # Odd length

    scrambled_list = list(binary_string)

    # Invert bit at every M-th position [cite: 228]
    # Note: Indexing assumes 1-based counting from the description, adjusted for 0-based
array
    for i in range(M - 1, T, M):
        current_bit = scrambled_list[i]

        # Bit Inversion
        if current_bit == '0':
            scrambled_list[i] = '1'
        else:
            scrambled_list[i] = '0'

    return "".join(scrambled_list)
```

A.2. Zero-Width Unicode Encoding

The following logic maps the scrambled binary data to the specific zero-width Unicode characters selected for this research.

Python

```
def encode_to_unicode(scrambled_binary):
    """
    Maps binary bits to zero-width characters[cite: 206, 207].
    """
    ZERO_WIDTH_0 = '\u288B'  # Represents Bit '0'
    ZERO_WIDTH_1 = '\u288C'  # Represents Bit '1'

    stego_string = ""

    for bit in scrambled_binary:
        if bit == '0':
            stego_string += ZERO_WIDTH_0
        elif bit == '1':
            stego_string += ZERO_WIDTH_1

    return stego_string
```

Appendix B: Cryptographic Core VHDL Specification

This appendix details the VHDL interface and Control Unit (FSM) logic for the AES-based JH hash function designed in Chapter 4.

B.1. Top-Level Entity Definition (JH_Wrapper)

The top-level entity defines the input and output ports used in the simulation test bench.

VHDL

```
-- Top Level Entity for the Cryptographic Core
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity JH_Wrapper is
    Port (
        clk      : in  STD_LOGIC;                      -- System Clock [cite: 297]
        rst      : in  STD_LOGIC;                      -- Reset Signal [cite: 297]
        mes_in   : in  STD_LOGIC_VECTOR(511 downto 0); -- Input Message Vector [cite: 298]
        hash_out : out STD_LOGIC_VECTOR(223 downto 0); -- Output Hash Digest [cite: 299]
        ack_out  : out STD_LOGIC                       -- Acknowledge/Done Signal [cite:
299]
    );
```

```
end JH_Wrapper;
```

B.2. Finite State Machine (FSM) Logic

The control unit is governed by a 5-state FSM as illustrated in the system architecture.

VHDL

```
-- FSM State Definitions
type state_type is (
    S0, -- Idle/Start: Initialize system [cite: 282]
    S1, -- First Round: Load initial message [cite: 283]
    S2, -- Iterative Rounds: Main loop (41 cycles) [cite: 284]
    S3, -- Check/Finalize: Finalize computation [cite: 287]
    S4, -- Process Next: Check for more blocks [cite: 288]
    S5  -- Done: Assert done signal [cite: 289]
);

signal current_state, next_state : state_type;
signal round_counter : integer range 0 to 42;

-- Process to handle state transitions
process(clk, rst)
begin
    if rst = '1' then
        current_state <= S0;
        round_counter <= 0;
    elsif rising_edge(clk) then
        current_state <= next_state;
        -- Counter logic for S2 loop
        if current_state = S2 then
            round_counter <= round_counter + 1;
        end if;
    end if;
end process;
```